

# Amey Walimbe

📍 Goleta, California    ✉ ameymwalmbe@gmail.com    ☎ 925-621-9646    🌐 Personal Website    in ameywalmbe  
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## Education

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### University of California, Santa Barbara

MS in Computer Science

BS in Computer Science

Expected June 2025

Sept 2020 – June 2024

- GPA: 3.81/4.0
- **Coursework:** Software Engineering, Data Structures and Algorithms, AI/Machine Learning, Probability, Mixed Reality, Computer Security, Computer Architecture

## Experience

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### Software Engineer Intern

Amazon Web Services

Seattle, WA

June 2024 – Sept 2024

- Spearheaded one-click load test campaign for AWS Batch, allowing developers to rapidly stress test and collect data for new features
- Integrated an automated load test approval step into the existing deployment process
- Collected and emitted common metrics (Batch job transition and TPS) during each load test
- Built a DynamoDB table and CloudWatch dashboard, offering real-time and historical performance insights, improving bottleneck detection

### Software Engineer Intern

Amazon

Sunnyvale, CA

Oct 2022 – Dec 2022

- Monitored 6 key FireTV metrics using AWS Cloud Services and visualized trends in Quicksight
- Processed millions of FireTV customer data points and constructed relevant datasets in Apache Spark
- Leveraged AWS Cloud Services such as Athena, S3, and Quicksight, and wrote comprehensive unit test cases
- Delivered weekly business reports with 10+ data visualizations, streamlining stakeholder decision-making
- Initiated migration of AWS Quicksight dashboard visualizations to Apache Superset

### Graduate Teaching Assistant

University of California, Santa Barbara

Goleta, CA

Oct 2024 – Dec 2024

- Mentored 100+ students in UCSB's Advanced Application Programming course, guiding them through REST APIs, the software engineering lifecycle, and end-to-end testing during class and office hours

## Projects

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### Towards Efficient and Robust Vision-Language-Action Models

[Report](#) [🔗](#)

- Replaced the LLaMA2-7B architecture in OpenVLA with TinyLlama-1B, then trained and finetuned on Llava-1.5 instruct, OpenX Embodiment, and LIBERO datasets
- Conducted systematic adversarial noise evaluations on OpenVLA, identifying failure points and proposing improvements to increase resilience against real-world perturbations
- *Built with OpenVLA, PrismaticVLM, PyTorch, LoRA, and Python*

### Bionic Vision Pipeline for Retinal Prosthesis Users

[Report](#) [🔗](#)

- Developed a context-aware salient object detection pipeline that integrates YOLOv8 for segmentation and GPT-4o for object saliency ranking and classification, enhancing bionic vision for retinal implant users
- *Built with YOLOv8, OpenCV, GPT-4o and Python*

### Exploring Synthetically Pre-Trained Depth Estimation Networks

[Published Paper](#) [🔗](#)

- Improved indoor/outdoor image classification accuracy by 4.4%, leveraging synthetically pre-trained depth-prediction networks to enhance generalization across diverse datasets.
- *Built with ResNet50 CNN, Omnidata, and Python*